

Name: _____

USN: _____

Course: Deep Learning (MMC321)

Total Marks: 50

Q1. Define the core principles of Deep Learning and explain key architectural layers. [2 Marks]

Deep Learning uses neural networks with multiple layers to learn useful representations from data. The main layers are input, hidden and output layers. Hidden layers learn features through weights, biases and activation functions.

Q2. Describe the fundamental workflow and mechanisms in Deep Learning. [2 Marks]

The workflow is: collect and preprocess data, choose a network, initialize parameters, perform forward propagation, calculate loss, use back-propagation to find gradients, and update parameters using an optimizer. This repeats for several epochs.

Q3. Explain key algorithms or modeling techniques used in Deep Learning with examples. [2 Marks]

Important techniques include back-propagation with gradient descent, CNNs for image recognition, and RNN/LSTM models for sequence data. A CNN can classify images by learning edges, shapes and higher-level features.

Q4. Analyze the transaction/query optimization and concurrency control in Deep Learning. [2 Marks]

These are primarily DBMS concepts rather than core Deep Learning concepts. In a DL system, similar concerns occur in data pipelines: concurrent data access must be controlled and database queries should be optimized for efficient data delivery.

Q5. Compare different design approaches in Deep Learning with their pros and cons. [2 Marks]

CNNs are effective for spatial data and learn local features efficiently. RNNs model sequences but may suffer from vanishing gradients. Transformers handle long-range dependencies well but can require substantial computation.

Q6. Solve a practical implementation scenario using Deep Learning best practices. [5 Marks]

For image classification, collect labelled images and split them into training, validation and test sets. Resize and normalize images and use augmentation when appropriate. Train a CNN using a suitable loss and optimizer such as Adam. Monitor validation performance to detect overfitting. Finally evaluate unseen data using accuracy, precision, recall and a confusion matrix.

Q7. Discuss security, integrity constraints, and error recovery in Deep Learning. [5 Marks]

Security includes protecting training data, model files, credentials and prediction APIs. Integrity means checking that input data is valid, consistent and not unintentionally corrupted. Error recovery can use checkpointing, logging, retries and restoring the latest valid model checkpoint. These are system-level concerns supporting reliable DL.

Q8. Design a scalable solution architecture for a real-world Deep Learning system. [5 Marks]

A scalable architecture can contain data ingestion, preprocessing, dataset storage, training services, a model registry and an inference API. Training can run on GPU workers and models can be versioned. Monitoring should track latency, errors and model performance. Load balancing and horizontal scaling support more users.

Q9. Define the core principles of Deep Learning and explain key architectural layers. [5 Marks]

Deep Learning uses neural networks with multiple learnable layers. The input layer receives features. Hidden layers transform them using weights, biases and activation functions such as ReLU. The output layer produces the prediction. During training, loss measures error, back-propagation calculates gradients and an optimizer updates the parameters.

Q10. Describe the fundamental workflow and mechanisms in Deep Learning. [10 Marks]

The workflow begins by defining the problem and collecting relevant data. Data is cleaned, transformed and divided into training, validation and test sets. A suitable neural-network architecture is then selected.

During forward propagation, inputs pass through the network and each layer transforms the representation using weights, biases and activation functions. The output is compared with the target using a loss function. Back-propagation applies the chain rule to calculate gradients, and an optimizer such as Adam updates the parameters.

Training is repeated over batches and epochs. Validation data is used to monitor generalization and detect overfitting. Regularization, dropout and data augmentation can improve generalization. The final model is evaluated on unseen test data using appropriate metrics and can then be deployed and monitored.

Q11. Explain key algorithms or modeling techniques used in Deep Learning with examples. [10 Marks]

Back-propagation computes gradients so an optimizer can update weights and minimize loss. Convolutional Neural Networks use convolution and pooling to learn spatial features and are widely used for image classification and computer vision.

Recurrent Neural Networks process sequential information and can be used for time series or language tasks. LSTM and GRU networks use gates to preserve useful information over longer sequences.

Multilayer perceptrons use fully connected layers and are useful for many tabular prediction tasks. Autoencoders learn compact representations by reconstructing their input. Transformer architectures use attention to model relationships between tokens and are widely used for language and other sequence problems.

The choice depends on the data and task: CNN for images, and LSTM or Transformer for sequential data. Performance should be verified using validation and test data.